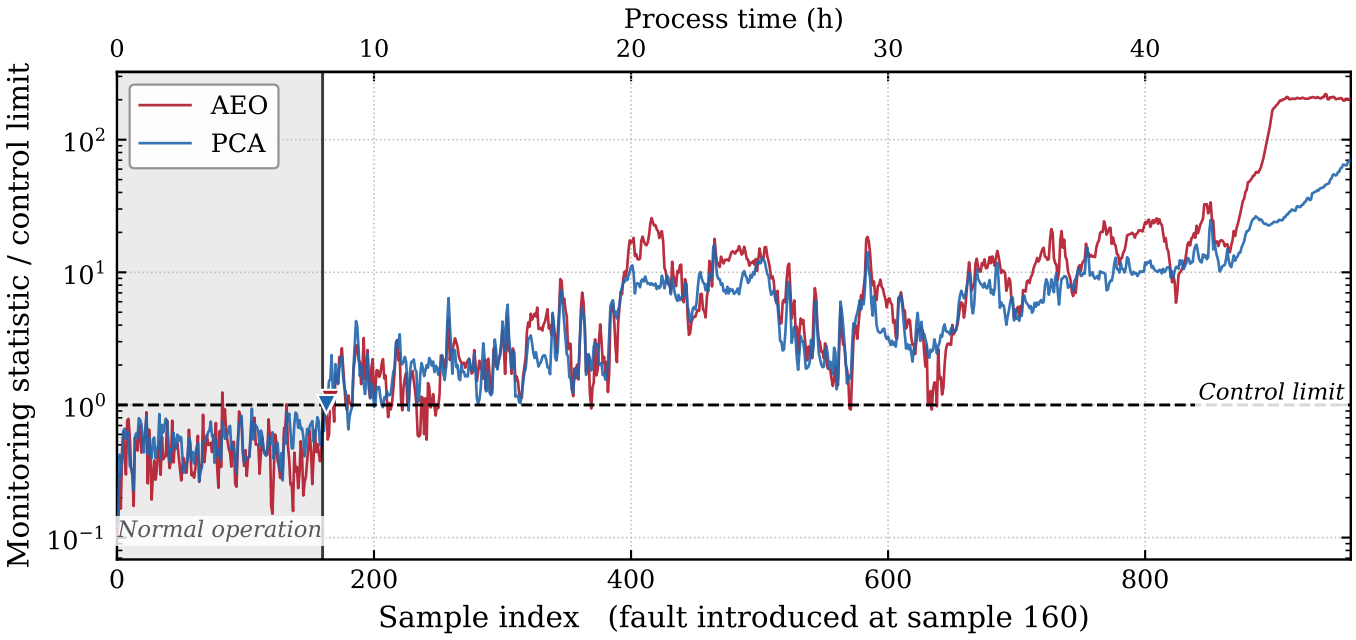
Fault Detection Performance for IDV(12)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	7.7 samples (23 min)	MDR 0.034	detected in 7/7 runs
PCA	delay	6.6 samples (20 min)	MDR 0.016	detected in 7/7 runs

PCA detects 1.1 samples (3 min) earlier; PCA missed-detection rate lower by 54%